

An Efficient Loan Approval Status Prediction Using Machine Learning

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Abstract— Banks derive a substantial portion of their profits from loans, making the loan approval process a vital undertaking for these financial institutions. However, it is becoming increasingly challenging to accurately assess loan requests and manage the risks associated with loan defaults. Due to this, scientists have been studying methods for forecasting loan approval systems, and machine learning techniques have proven to be effective in handling large amounts of data. In this study, we utilize algorithms such as K-Nearest Neighbours and Decision Trees to predict loan approval for customers. We also introduce a novel algorithm, the Support Vector Classifier, which demonstrates superior accuracy and is used to deploy the model. With these machine learning techniques, we are able to develop a bank loan prediction system that automatically selects eligible candidates for loan approval. The research study explored the application of machine learning algorithms in predicting the status of loans. SVM was one of the algorithms evaluated, and it achieved an accuracy score of 83%. Nonetheless, the research also revealed that various factors can impact the efficacy of SVM, such as the quality of the data, the choice of hyperparameters and the existence of data bases in the data or model. Overall, the study found that SVM has the potential to be a valuable tool for loan status prediction, but its success will depend on a number of factors.

Keywords— *Loan approval, Loan Default, Loan prediction, Machine learning, Support vector machine.*

I. INTRODUCTION

Banks primarily rely on loans as their main income stream while also exposing themselves to financial vulnerability. The interest earned from loans is a major contributor to a bank's assets [2]. When lending loans, a major risk known as "credit risk" arises due to the borrower's capacity to repay the loan on time as per the given timeframe. This factor cannot be overlooked. [10]. The eligibility of a loan applicant for approval or denial was based on a credit score, which is a numerical score. The study aims to explain the use of different machine learning methodologies that can accurately determine the appropriate candidates for loan approval and help banks recognize defaulters to reduce credit risk substantially. [11] Banks use datasets to evaluate loan applicants and classify them into

good or bad classes based on their likelihood of returning the money borrowed [9]. Good- class applicants are more likely to repay the loan while bad- class applicants have a higher probability of default. However, human decision-making is influenced by non-quantifiable factors such as misjudgment, political changes, accidents, outbreaks, and unpredictable weather, making accurate default prediction models necessary[8]. It is a complex task for banks to assess the nature of a customer's adherence to banking rules[12][20]. Banks usually categorize customers as good or bad based on their ability to pay monthly bills and repay loans on time [18]. Machine learning algorithms are employed to create models that accurately classify loan applicants based on their repayment potential [15].

Managing and evaluating credit risk poses a vital challenge within the banking sector due to its complexity. Banks must involve different departments or external parties in their approval processes, which can be time-consuming and may cause them to lose business opportunities. The main focus of banks is to distribute loans, and they require a system that is both impartial and streamlined in order to save time and avoid bias when assessing loan applications [16]. To accurately group data items into their respective categories, a method called classification system is utilized. A model is developed using a set of training data which helps in the accurate classification of the data items. No information has been left out while paraphrasing the original text. To ascertain a client's ability to repay a loan, a test dataset is generated [17].

II. RELATED WORKS

Bhatt, Kusumlata, et al. [1] proposed a model utilizing machine learning models with Ensemble Learning Techniques to determine the viability of granting individual loan requests. By using this approach, it is possible to enhance the accuracy with which suitable candidates are selected from an existing list.

Mohammad Ahmad Sheikh et al. [3] proposed model aims to predict whether a bank will grant a loan to its customers. The model targets classification and uses Logistic

Regression with a sigmoid function for its development. Preprocessing is a major area of the model, followed by exploratory data analysis, feature engineering, and model selection. The model is fed with two separate datasets and then proceeded with.

Wenyu Qiu et al. [7] developed a two-channel credit evaluation model with a shunt. This model aims to accelerate the evaluation process for customers with positive behavior, while penalizing those with negative behavior. The goal is to create a more efficient system for evaluating creditworthiness. Empirical tests have shown that this model can simplify the overall evaluation process, accelerate calculations, and improve model indicators. Moreover, by configuring multiple parameters sets in the "shunt," it can automatically regulate the loan approval rate based on various loan policies.

Anjali Kulkarni et al. [19] utilized various models including Logistic Regression, SVM and Decision Tree were explored for their effectiveness in categorizing loan requests. Results from analyzing these models with a typical data set indicated that the Multilayer Perceptron Model and Logistic Regression could handle missing data and still perform efficiently. Therefore, the objective is to create an integrated model that leverages the strengths of all five models.

Jay Lohokare et al. [13] have proposed a system in their system that simplifies data collection by consolidating all user-related data into a single application. With this system, there is no longer a reliance on external institutions or banks to access such data. They also proposed an ANN (Artificial Neural Network) based model to determine the final loan status.

G. Arutjothi et al. [14] have suggested a loan status model that can distinguish between valid and default customers among loan applicants. The proposed model achieved 75.08% accuracy in classifying credit applicants using the R package. Credit lenders can use this model to determine loan proposals. Additionally, a comparative study was conducted using various iteration levels, which found that the 30th iteration- based k-NN model was more accurate than other levels. This model can aid in preventing substantial commercial bank losses.

Vishal Singh et al. [2] have identified that in certain situations, such as a client undergoing a disaster, the algorithm may not be able to provide an accurate prediction. However, their research paper can still identify potential clients with a good accuracy rate for loan repayment. Factors such as loan duration, amount, age, and income play a significant role in determining a client's potential for loan repayment. On the other hand, "zip code" and "credit history" are crucial factors in predicting the loan applicant's category. To predict loans, they have proposed an XGBoost model.

Anshika Gupta et al. [5] emphasize the importance of discovering new technology and updating old ones to minimize human intervention and increase work efficiency in today's rapidly growing IT industry. To predict loan approval status for banking systems or loan applicants, they have implemented a Random Forest model that proves to be highly useful in bank management. The data analysis results indicate that this model reduces fraudulent activity that may occur during loan approval.

Mr. R. Karthiban et al. [6] concentrated on solving a classification issue and employed the Naive Bayes technique to categorize information based on the activation function. The choice of an appropriate algorithm depends on the nature of the target feature, which could be categorical or discrete (countable values) or continuous (values that change with time). In the case of bank loan prediction, the two possible outcomes are loan allocation or no loan allocation, which is categorized under the two-class classification problem. The authors explored various classification algorithms for predicting bank loans and presented a dataset with its features and characteristics as an example.

Abhishek Shivanna et al. [4] used ML models such as AP, DSVM, BPM and BDT algorithms on Azure Machine Learning platform to predict credit card defaulters with high accuracy. The evaluation metrics are presented in tables and DSVM outperformed all other models with an 82.2% overall accuracy and a 0.74 AUC. DSVM showed great potential to be used in defaulter detection systems. In the future, the team plans to train a defaulter detection system using Multilayer Perceptron.

Based on the related works inputs were taken to incorporate an efficient machine learning algorithm to provide accurate prediction results for estimating the loan approval status of an individual.

III. PROPOSED METHODOLOGY

A. Dataset Description:

The problem utilizes a dataset obtained from Kaggle, which is free of any missing information. The dataset pertains to loan approvals and consists of 615 rows of applicant data, all of which are valid. The loan approval status is provided for each applicant based on various attributes.

TABLE. I FACTORS TAKEN INTO CONSIDERATION FOR LOAN APPROVAL STATUS PREDICTION

S.no.	Feature Name	Description	Value
1.	Gender	Gender of the applicant	Male or Female
2.	Married	Marital status	Yes or No
3.	Education	Education of the customer	Graduate or Not a Graduate
4.	Applicant Income	Income	Integer value
5.	Co applicant income	Income	Integer value
6.	Property area	Area of residence	Urban Semi-urban Rural
7.	Loan status	Approval status of the loan	Yes or No

B. Data Pre-processing:

To perform data normalization, the information is transformed into a particular type, such as float or double,

through a process known as type casting, which involves converting a value into a different format.

This technique is used to convert categorical values, such as gender, marital status, education, and loan status, into numerical values to enhance the accuracy and effectiveness of model evaluation. No information is left out during this process.

C. Feature Engineering:

The act of determining the most effective characteristics that positively impact the forecasted outcome is known as Feature Engineering. To accomplish Feature Selection, we utilized the Correlation Matrix. It is essential to consider all relevant information when conducting this process.

Correlation Matrix:

The Correlation matrix is a table that displays the correlation among different features and target variables. It utilizes Pearson's correlation coefficient, which is calculated through a particular formula. The table must show all the correlations between the variables.

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}} \text{-----(1)}$$

The Quantity of Information is represented by the variable n, while the first variable value is represented by $\sum x$ and the second variable value is represented by $\sum y$. The product of the first and second value is represented by $\sum xy$, and $\sum x^2$ and $\sum y^2$ represent the sum of the squares of the first and second values, respectively. All of these variables are essential in determining the necessary information.

Correlation Matrix for features and target:

In a dataset where n=615 and the X values include gender, marital status, and education while the Y value is the status of a loan, the correlation matrix revealed a negative correlation between the 'region' feature and charges. Therefore, the feature was removed as it did not impact the target variable. While some other features had a lesser impact on the target variable, they were still included in the analysis as they were not as significant as the 'region' feature.

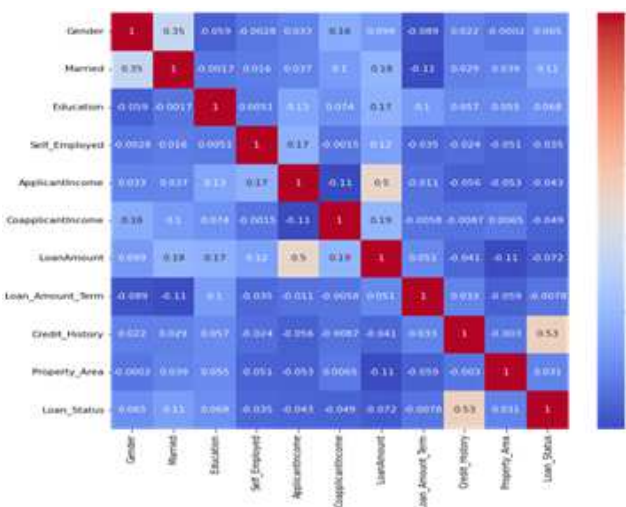


Fig. 1. Correlation matrix

D. Splitting the input data:

To avoid bias, we followed the Pareto Principle and randomly divided the input data into two sets: The dataset was divided into two parts, where the first part consisted of 80% of the data and was used for training, while the second part consisted of the remaining 20% and was reserved for testing. The most efficient support vector machine model was used to train the model on the training set and evaluate it against the test set.

Support vector machines are supervised Algorithms for machine learning that are utilized for the purposes of analyzing regression and classification.

The primary goal of SVM is to locate the best possible hyperplane that can efficiently distinguish between data points from various categories, or precisely forecast the continuous value of a desired outcome, using the concept of maximum margin.

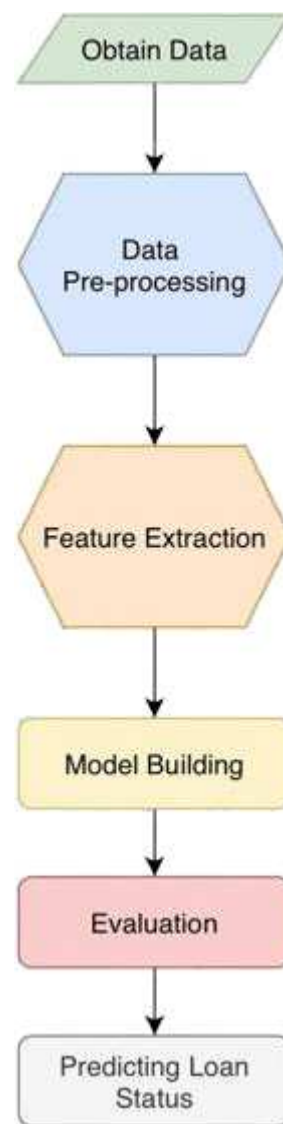


Fig. 2. Working Methodology

The support vectors, which are the closest data points of different classes, are used to determine the hyperplane that

maximizes the margin in SVM. SVM utilizes a technique of transforming the input data into a space of higher dimensions to segregate the data more effectively.

The conversion process takes place by employing a kernel function that effectively transforms the data into a newly defined space. SVM (Support Vector Machine) stands out as a widely used machine learning algorithm due to its ability to handle high-dimensional data, effectively address non-linearly separable data, and prevent overfitting.

Our SVM model as given in fig3 consists of training and testing process, the training process involves finding the dataset and computing the hyperplane that divides training data and test data, while the testing process is evaluating the trained model's performance on new data.

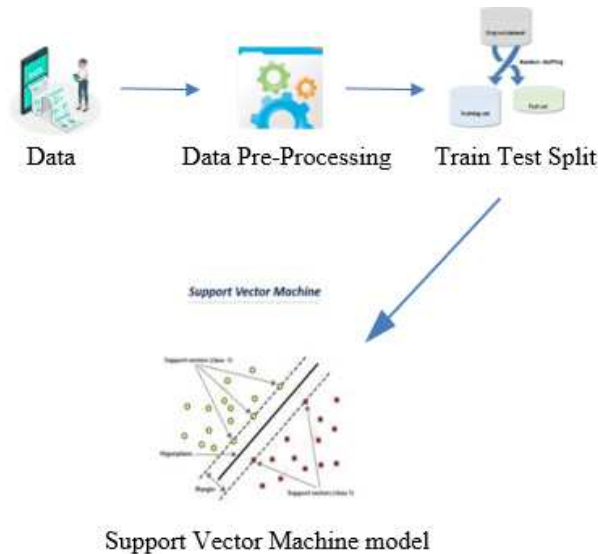


Fig. 3. Model Architecture

IV. RESULTS AND DISCUSSION

To determine if an individual is eligible for a loan, a support vector classifier algorithm was utilized on various features such as gender, marital status, number of dependents, self-employment status, applicant income, co-applicant income, loan amount, credit history, and property location. Based on validation and testing, the model has achieved accuracy score of 83%. This model was implemented on a 12 GB RAM machine with 107 GB storage. The model developed is inexpensive and accessible to all. We have also hosted a website <https://loanstatusprediction.streamlit.app/> for individuals to get an idea about them being qualified on approval for a loan. When considering factors like gender, marital status, dependents, self-employment, applicant income, co-applicant income, loan amount, loan term, and credit history, Property area where the categorical and non-categorical Data are converted to float values on which support vector classifier is applied and their chance of being Approved for a loan will be predicted. We have run various algorithms based on data obtained from Kaggle and have got a better accuracy score.

The screenshot shows a web form for loan status prediction. The form includes the following fields and values:

- Gender: Male
- Married: Yes
- Dependents: 2
- Education: Graduate
- Self Employed: Yes
- Applicant Income: 50000
- Co Applicant Income: 25000
- Loan Amount: 2000000
- Loan_Amount_Term: 360
- Credit_History: 1
- Property_area: Urban
- Loan Status: Loan approved

Fig. 4. Website Screenshot

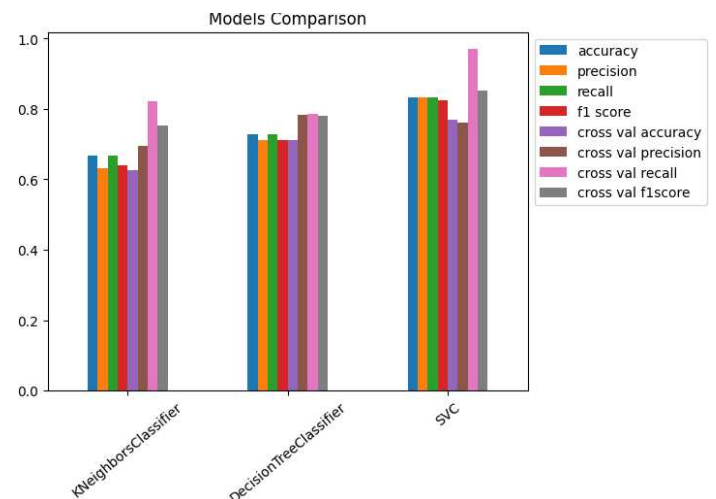


Fig. 5. Model Comparison

Our study entails a comprehensive comparative analysis between our proposed ML models and the existing models in order to evaluate their performance, effectiveness, and suitability for the given task.

Table II shows the values obtained from the existing models for prediction.

TABLE. II VALUES OBTAINED FROM EXISTING MODEL.

	Accuracy	Precision	Recall	f1 score
Two class Bayes Point Machine	80.60%	0.73	0.2	0.31
Two class decision tree	81.60%	0.62	0.45	0.52
Two class avg perception	80.90%	0.71	0.24	0.36

Table III shows the values that are obtained from our proposed system.

TABLE III. VALUES OBTAINED FROM OUR PROPOSED MODELS

Metrics	K Neighbors Classifier	Decision Tree Classifier	SVC
Accuracy	0.666667	0.729167	0.833333
Precision	0.632479	0.712171	0.831695
Recall	0.666667	0.729167	0.833333
f1 score	0.638889	0.71162	0.825275
Cross val accuracy	0.627159	0.710532	0.768458
Cross val precision	0.694547	0.783135	0.761696
Cross val recall	0.822712	0.785819	0.97
Cross val f1 score	0.75252	0.779391	0.85306

V. CONCLUSION

The Research focuses on machine learning algorithms that can be used for effective loan status prediction along with support vector classifier which helps to prioritize the parameters used for prediction. The support vector classifier faired among the models with an accuracy score of 83%. To draw a conclusion on the effectiveness of the SVM model for this specific problem, it would be necessary to evaluate its performance metrics, such as accuracy, precision, recall, and F1 score, on a test set of data. Additionally, when training a model, it is crucial to take into account various aspects, including but not limited to the amount and accuracy of data used, selecting appropriate hyperparameters, and identifying any possible biases in both the data and model. In general, SVMs have the potential to be a strong resource for forecasting loan approval status. However, their success will be influenced by numerous factors and must be assessed on an individual basis. As more data becomes available and computational resources advance, SVM models can be continuously trained and fine-tuned to improve their predictive accuracy. Enhancing the model's performance and adaptability to dynamic market conditions can be achieved by integrating new features, economic indicators, or alternative data sources.

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